

Linear regression – Interpretation

- *what is Regression*
- *What is Linear regression*
- *What is Simple linear regression*
- *What is Multiple linear regression*
- *Importance of coefficients*
- *Importance of slope and bias*
- *Error*
- *SSE*
- *MSE*
- *RMSE*
- *Loss vs Cost*
- *Cost function*
- *Goal*
 - *How to find the suitable coeff in order to minimize the cf*
- *OLS*
- *We found equation of Coeff === (model is ready)*

Start prepare the document

Presentation session on LR on next Saturday

$$Y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n$$

$$\text{Sales} = b_0 + b_1 * TV + b_2 * SM + b_3 * NP$$

Problem statement

We are spending money on TV, SM and NP how much your sales are increasing

Model Equation

$$Sales = b_0 + b_1 * TV + b_2 * SM + b_3 * NP$$

Solution:

by using OLS (ordinary least square) method we are finding the coefficients

Assume that $b_0 = 1000, b_1 = 20, b_2 = 30, b_3 = 10$

$$Sales = 1000 + 20 * TV + 30 * SM + 10 * NP$$

Model is created, now you are showing to the client what is the meaning of this equation

Interpretation of Linear regression

$$Sales = 1000 + 20 * TV + 30 * SM + 10 * NP$$

you invest more on SM ..then on TV

you want focus more on SM

$$y = 2x, y = 3x, y = 4x$$

x	$y = 2x$	$y = 3x$	$y = 4x$	$Y = bx$
1	2	3	4	
2	4	6	8	
3	6	9	12	
4	8	12	16	
5	10	15	20	
6	12	18	24	

$$Y = b_0 + b_1 * x$$

Interpretation of b_1 :

If we increase x by one unit , then y will be increase by b_1 times

Keeping other coefficients as constant

$$\text{Sales} = 1000 + 20 * TV + 30 * SM + 10 * NP$$

If we increase TV by one unit , then Sales will be increase by 20 times

Keeping other coefficients as constant

If we increase SM by one unit , then Sales will be increase by 30 times

Keeping other coefficients as constant

If we increase NP by one unit , then Sales will be increase by 10 times

Keeping other coefficients as constant

$$Y = b_0 + b_1 * x$$

Interpretation of b_0 :

$$\text{Sales} = 1000 + 20 * TV + 30 * SM + 10 * NP$$

If we don't spend any thing on SM , NP and TV still we have 1000 sales

*Bias will give the indication how much constant we can add
even though when we dont have our data*

the interpretation possible because of Linearity

Non linear equation can not be interpretable